
Notes de stage

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1 Expression du reste

1.1 Expression exacte du reste

On veut trouver une expression de $\frac{1}{|C_i|} \int_{C_i} \nabla u$ à partir des u_k :

$$\frac{1}{|C_i|} \int_{C_i} \nabla u = \frac{1}{|C_i|} \int_{\partial C_i} u(x) n(x) dx = \frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} \left[\int_a^b u(x^{i,j}(\xi)) d\xi \right] n_{i,j} \quad (1)$$

Où $x^{i,j} : [a, b] \rightarrow \Gamma_{i,j}$ est une paramétrisation de $\Gamma_{i,j}$.

Effectuons un d.l. :

$$u(x) = u(x_{i,j}) + [x - x_{i,j}] \cdot \nabla u(x_{i,j}) + \int_0^1 (1-t)(x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dt$$

où $\gamma_x(t) = tx_{i,j} + (1-t)x$.

$$\text{Ainsi : } u_k = \frac{1}{|C_k|} \int_{C_k} u = u(x_{i,j}) + [x_k - x_{i,j}] \cdot \nabla u(x_{i,j}) + \int_0^1 \frac{(1-t)}{C_k} \int_{C_k} (x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dx dt$$

Pour $k \in \{i, j\}$.

On a ainsi :

$$\frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} = u(x_{i,j}) + \int_0^1 \frac{(1-t)}{d_{i,j}} \left[\frac{d_{j,ij}}{C_i} \int_{C_i} (x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dx + \frac{d_{i,ij}}{C_j} \int_{C_j} (x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dx \right] dt$$

On pose :

$$R_1^{i,j} = \int_0^1 \frac{(1-t)}{d_{i,j}} \left[\frac{d_{j,ij}}{C_i} \int_{C_i} (x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dx + \frac{d_{i,ij}}{C_j} \int_{C_j} (x - x_{i,j})^t H_{\gamma_x(t)} u(x - x_{i,j}) dx \right] dt \quad (2)$$

Ainsi :

$$\frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} = u(x_{i,j}) + R_1^{i,j} \quad (3)$$

Effectuons cette fois un d.l. sur $x(\xi)$:

$$u(x^{i,j}(\xi)) = u(x_{i,j}) + [x^{i,j}(\xi) - x_{i,j}] \cdot \nabla u(x_{i,j}) + \int_0^1 (1-t)(x^{i,j}(\xi) - x_{i,j})^t H_{\gamma_{x^{i,j}(\xi)}(t)} u(x^{i,j}(\xi) - x_{i,j}) dt$$

Ainsi, en utilisant (3) on a :

$$u(x^{i,j}(\xi)) = \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} + [x^{i,j}(\xi) - x_{i,j}] \cdot \nabla u(x_{i,j}) + \int_0^1 (1-t)(x^{i,j}(\xi) - x_{i,j})^t H_{\gamma_{x^{i,j}(\xi)}(t)} u(x^{i,j}(\xi) - x_{i,j}) dt - R_1^{i,j}$$

Et en intégrant :

$$\int_a^b u(x^{i,j}(\xi))d\xi = \lambda_{i,j} \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} + \lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \nabla u(x_{i,j}) + \int_0^1 (1-t) \int_a^b (x^{i,j}(\xi) - x_{i,j})^t H_{\gamma_{x^{i,j}}(\xi)}(t) u(x^{i,j}(\xi) - x_{i,j}) d\xi dt - R_1^{i,j}$$

Où $\lambda_{i,j} = |\Gamma_{i,j}|$.

On pose :

$$R_2^{i,j} = \int_0^1 (1-t) \int_a^b (x^{i,j}(\xi) - x_{i,j})^t H_{\gamma_{x^{i,j}}(\xi)}(t) u(x^{i,j}(\xi) - x_{i,j}) d\xi dt \quad (4)$$

Ainsi :

$$\int_a^b u(x^{i,j}(\xi))d\xi = \lambda_{i,j} \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} + \lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \nabla u(x_{i,j}) + R_2^{i,j} - R_1^{i,j} \quad (5)$$

Effectuons un dernier d.l. :

$$\frac{1}{|C_i|} \int_{C_i} \nabla u = \frac{1}{|C_i|} \int_{C_i} \left[\nabla u(x_{i,j}) + \int_0^1 (1-t) H_{\gamma_x(t)} u \times x dt \right] dx$$

$$= \nabla u(x_{i,j}) + \int_0^1 \frac{1-t}{|C_i|} \int_{C_i} H_{\gamma_x(t)} u \times x dx$$

Ainsi :

$$\frac{1}{|C_i|} \int_{C_i} \nabla u = \nabla u(x_{i,j}) + \int_0^1 \frac{1-t}{|C_i|} \int_{C_i} H_{\gamma_x(t)} u \times x dx \quad (6)$$

Et en injectant (6) dans (5) on obtient :

$$\int_a^b u(x^{i,j}(\xi))d\xi = \lambda_{i,j} \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} + \lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \left(\frac{1}{|C_i|} \int_{C_i} \nabla u - \lambda_{i,j} \int_0^1 \frac{1-t}{|C_i|} \int_{C_i} H_{\gamma_x(t)} u \times x dx \right) + R_2^{i,j} - R_1^{i,j}$$

On pose :

$$R_3^{i,j} = \lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \int_0^1 \frac{1-t}{|C_i|} \int_{C_i} H_{\gamma_x(t)} u \times x dx \quad (7)$$

Ce qui permet d'écrire :

$$\int_a^b u(x^{i,j}(\xi))d\xi = \lambda_{i,j} \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} + \lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \frac{1}{|C_i|} \int_{C_i} \nabla u + R_2^{i,j} - R_1^{i,j} - R_3^{i,j} \quad (8)$$

Finalement en injectant (8) dans (1), on obtient :

$$\frac{1}{|C_i|} \int_{C_i} \nabla u = \frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} \lambda_{i,j} \frac{d_{j,ij}u_i + d_{i,ij}u_j}{d_{i,j}} n_{i,j} + \left[\lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \frac{1}{|C_i|} \int_{C_i} \nabla u \right] n_{i,j} + [R_2^{i,j} - R_1^{i,j} - R_3^{i,j}] n_{i,j} \quad (9)$$

Et on pose :

$$R_4^{(i)} = \frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} [R_2^{i,j} - R_1^{i,j} - R_3^{i,j}] n_{i,j} \quad (10)$$

Ce qui permet d'écrire :

$$\overline{\nabla u(x_i)} = \frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} \left[\lambda_{i,j} \frac{d_{j,ij} u_i + d_{i,ij} u_j}{d_{i,j}} n_{i,j} + \left[\lambda_{i,j} (\bar{x}_{i,j} - x_{i,j}) \cdot \overline{\nabla u(x_i)} \right] n_{i,j} \right] + R_4^{(i)} \quad (11)$$

$$\text{où } \overline{\nabla u(x_i)} = \frac{1}{|C_i|} \int_{C_i} \nabla u.$$

On peut le reformuler de la manière suivante :

$$B_i \overline{\nabla u(x_i)} = \left(\frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} \lambda_{i,j} \frac{d_{j,ij} u_i + d_{i,ij} u_j}{d_{i,j}} n_{i,j} \right) + R_4^{(i)} \quad (12)$$

$$\text{où } B_i = I_2 - \frac{1}{|C_i|} \sum_{j \in \mathcal{N}_i} \lambda_{i,j} n_{i,j} \otimes (\bar{x}_{i,j} - x_{i,j})$$